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Neutrophil-derived exosome from systemic sclerosis inhibits the proliferation and migration of endothelial cells

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ABSTRACT

Objectives: Systemic sclerosis (SSc) is an autoimmune disease characterized by vasculopathy, inflammation, and extensive fibrosis in multiple organs. Exosomes (EXOs) are cell-derived vesicles contained various DNAs, RNAs and proteins, and play important roles in various diseases. Here, we aimed to investigate the roles of SSc EXOs in angiogenesis related mechanisms.
Methods: EXOs were isolated from plasma, cultured peripheral blood mononuclear cells (PBMCs)/neutrophil supernatants, and identified by transmission electron microscopy. The expression of S100A8/A9 was measured by real-time PCR and ELISA. Proliferation, migration and scratch assays in human dermal microvascular endothelial cells (HDMECs) were used to study the EXOs influence.
Results: Plasma and neutrophil EXOs from SSc patients can suppress the proliferation and migration of HDMECs. High levels of S100A8/A9 were found in SSc EXOs which derived from plasma, PBMCs and neutrophils. The expression of S100A8/A9 in neutrophil EXOs was higher than that in PBMC EXOs in SSc patients. The proliferation and migration of HDMECs were possibly inhibited by S100A8/A9 of neutrophil EXOs.
Conclusions: Neutrophil EXOs from SSc patients inhibits the proliferation and migration of HDMECs, S100A8/A9 might play an important role in this process.

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1. Background

Systemic sclerosis (SSc) is an autoimmune disease of unknown aetiology characterized by vasculopathy, inflammation, and extensive fibrosis of skin and organs [1], which has the highest mortality among connective tissue diseases [2]. There is no effective anti-fibrosis drug in SSc currently. The pathogenesis of SSc is complicated, and vascular endothelial damage is considered as the initiating factor [3]. The vasculopathy caused by vascular remodelling and abnormal neovascularization includes destructive lesions and hyperplasia occlusive lesions, which can result in a progressive decrease in the capillary bed, arterial occlusion and fibrosis, increasing neovascular permeability. Finally, the vasculopathy could lead to varied clinical manifestations, such as pulmonary arterial hypertension, scleroderma renal crisis, fingertip ulcers, Raynaud's phenomenon and so on [4]. Therefore, the search for specific interventions and effective treatment targets for SSc-related fibrosis is likely to concentrate on vascular structures and functional disorders.

Exosomes (EXOs) are 30–150 nm cell-derived extracellular spherical vesicles, they contain a complex cargo of various DNAs, RNAs and proteins [5]. The characterized markers of EXOs include annexins, flotillin, integrins and tetraspanins, such as CD63, CD9, and CD81/82. EXOs are present in all biological fluids, such as serum, plasma, and can be secreted by all cells including epithelial cells, fibroblasts, and immune cells. EXOs derived from different cells have different compositions, and the functional molecules secreted in EXOs have greater bioactivity than those in their soluble form [6].

EXOs participate in intercellular signalling, mainly acting in processes such as information transfer, inflammation, antigen presentation, tumour growth, metastasis, and angiogenesis [5]. EXOs were widely studied in various diseases. For example, EXO-derived miRNAs act as predictive biomarkers for diffuse large B-

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cell lymphoma chemotherapy resistance [7]. EXO-transmitted lncRNA activated in RCC with sunitinib resistance (lncARSR) promotes sunitinib resistance in renal cancer by acting as a competing endogenous RNA [8]. In rheumatology diseases, EXOs act as carriers of autoantigens and cytokines [9], with EXO-derived Hotair acting as a critical regulator and potent marker for rheumatoid arthritis [10] and urine EXOs in systemic lupus erythematosus acting as potential renal disease markers [11]. In SSc, there are six profibrotic miRNAs increased and ten antifibrotic miRNA decreased in serum EXOs, which induced a profibrotic phenotype in cultured normal dermal fibroblasts [12]. Elevated plasma endothelial microparticles in SSc patients were negatively correlated with severity of skin involvement and capillary loss by nailfold video-capillaroscopy [13]. Serum EXOs levels were also negatively correlated with skin ulcers or pitting scars and the wound-healing delay [14].

Here, we firstly studied the influence of SSc EXOs on the proliferation and migration of human dermal microvascular endothelial cells (HDMECs). As S100A8/A9 was reported to locate on the surface of EXOs, and our previous microarray analysis indicated S100A8/A9 might be secreted with EXOs form. Then we studied the function of EXOs S100A8/A9 in the vascular pathology of SSc.

2. Materials and methods

2.1. Patients & controls

Whole blood samples were obtained from 39 diffuse cutaneous SSc patients and 32 healthy individuals from Xiangya Hospital of Central South University in this study. All patients and normal controls (NC) were Han Chinese and signed informed consents, which was approved by the Ethical Committee of Xiangya Hospital of Central South University (No. 201404360). All SSc patients fulfilled the American College of Rheumatology classification criteria for SSc [15]. Among all patients, 19 SSc patients and 18 age- and gender-matched normal controls were included for the mRNA transcription profiling (As describe previously [16]); the other 20 SSc patients and 14 NC were included to validate mRNA expression and EXO section in this study. The details of the SSc patients and NC are shown in the supplementary material [Table S1].

2.2. Isolation of plasma, neutrophils and PBMCs & cell culture

For the EXO study, 10 ml-peripheral whole blood samples were collected in preservative-free EDTA anticoagulant tubes from patients and controls and processed within 2 h. Plasma, neutrophils and PBMCs were isolated by density gradient equilibrium centrifugation using Histopaque-1077 and Histopaque-1119 (Sigma-Aldrich, St. Louis, USA) according to the manufacturer's instructions. After centrifugation at 800×g for 40 min at room temperature, six distinct layers were observed: plasma (light yellow transparent layer), peripheral blood mononuclear cells (PBMCs) (opaque layer), Histopaque-1077, neutrophils (opaque layer), Histopaque-1119 and erythrocytes. The plasma, PBMCs and neutrophils were transferred to separate tubes, and plasma was stored at −80 °C until analysis.

The cells (cell viability>90%) were resuspended in the appropriate volume (2−5x10^6 cell/ml) of 1640 media (Gibco, USA) supplemented with 10% EXO-free foetal bovine serum (FBS, Gibco, USA, prepared by obtaining the supernatant after centrifugation at 150,000×g for 3 h at 4 °C). Neutrophils were cultured for 2 h, and PBMCs were cultured for 24 h at 37 °C in a 5% CO2 humidified incubator. The cell pellets and supernatants were all collected for further analysis.

For mRNA transcription profiling, PBMCs were isolated from whole blood through density gradient centrifugation using Ficoll-Pague PLUS (GE Healthcare, Life Sciences, UK) in accordance with the manufacturer's instructions.

2.3. Isolation & identification of EXOs

Due to the limited volume of samples available from patients, we isolated EXOs from plasma (plasma EXOs) and PBMC/neutrophil cultured supernatants (PBMC/neutrophil EXOs) using an exoEasy Maxi Kit (Qiagen, Frederick, USA) according to the manufacturer's instructions. Prefiltered plasma (4ml/sample) or cell cultured supernatants (20−30ml/sample) finally collected 600 μl EXOs buffer and stored at −80 °C until analysis.

EXOs were identified by size distribution using dynamic light scattering (DLS) on a Zetasizer Nano ZS (Malvern Instrument, UK) and by morphology using transmission electron microscopy (TEM) (FEI Tecnai G2 Spirit Twin, Czech Republic) via negative staining. We also performed bead-based flow cytometry with Exosome-Human CD63 Isolation/Detection Dynabeads (Invitrogen, Thermo Fisher Scientific, Lithuania) and an anti-human CD63-PE (12−0639, eBioscience, USA), an anti-S100A8/A9 antibody (ab17050, Abacm, UK) and FITC-conjugated goat anti-mouse IgG (H + L) (Jackson ImmunoResearch, West Grove, USA) as secondary antibodies according to the manufacturer's protocol. We used a FACSCalibur flow cytometer (BD Biosciences, USA) and FlowJo software (TreeStar) for analysis.

2.4. RNA isolation & real-time quantitative PCR

Total RNA was isolated from plasma EXOs and PBMCs using Trizol (Invitrogen Life Technologies, CA, USA), and cDNA was obtained from a GoScript Reverse Transcription System (Promega, USA). The relative expression of genes was measured by gene-specific primers [shown in Supplementary Table 2S] with SYBR Green (SYBR Premix Ex Taq RT-PCR kit, Takara, Japan) and the 7500HT Fast Real-Time PCR system (Applied Biosystems, CA, USA). The housekeeping gene β-ACTIN was selected for normalizing and calculating the relative quantity (RQ) of gene expression in each sample.

2.5. Cell culture & treatment

HDMECs purchased from CellBio (catalogue # CBR130858) [17], and were cultured in Dulbecco's modified Eagle's medium (DMEM, Gibco, USA) supplemented with 10% EXO-free FBS at 37 °C in a 5% CO2 humidified incubator. The cells were used when they reached 80−90% confluence.

2.6. HDMEC proliferation assay

HDMECs were subjected to a 7Sea-Cell Counting Kit (7Sea Pharmatech, China), and cell metabolic activity/proliferation was assessed using manufacturer-recommended protocols. Then, 100 μl of HDMECs (2×10^4 cells/ml) was added into each well of 96-well plates and allowed to adhere completely at 37 °C in a 5% CO2 humidified incubator. The supernatants were transferred to medium with 2.5% plasma/neutrophil EXOs, 100 ng/ml S100A8 protein (ab167749, Abacm, UK) and 500 ng/ml S100A8/A9 protein (8916-S8, R&D Systems, USA). At 0, 3, 6, 12, and 24 h, 10 μl of MTS reagent was added to each well for 2 h. Then, absorbance was read at 450 nm in a microplate reader (BioTek, Winooski, VT). The optical density value was corrected relative to the number of cells. Each independent experiment was performed three times with three duplicates in each group.

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2.7. HDMEC transwell migration assay

We used 8-μm-pore-size transwell migration chambers (Costar, Corning, NY) for the migration analysis with neutrophil EXOs from SSc patients and NC, as well as the S100A8 antibody (AF 4570, R&D Systems, USA) and S100A8 protein. A total of 100 μl of HDMECs (2×10^6 cells/ml) was plated in the top of the chamber, and 600 μl DMEM supplemented with 10% EXO-free FBS, 2.5% neutrophil EXOs from SSc patients and NC, 100 ng/ml S100A8 protein, and 1.25 ng/ml S100A8 antibody was added to each well. After the cells were cultured for 24 h, they were fixed with ethanol for 30 min. The cells remaining at the top of the chamber were removed, and the cells that adhered to the bottom of the chamber were stained with toluidine blue. We counted the cells using a phase-contract microscope. Each independent experiment was performed three times.

2.8. HDMEC scratch assay

HDMECs were grown to confluency in six-well culture plates. A pipette tip was used to scratch a cross-shaped wound uniformly placed in the well centre. Wells were washed with PBS, and DMEM was added with or without 2.5% plasma/neutrophil EXOs. The demarcated areas of each well were recorded at 0, 12, and 24 h under a phase contrast microscope using Improvision software. We analysed pictures using NIH ImageJ software to calculate the wound area. Each independent experiment was performed three times in 2–3 duplicates in each group.

2.9. S100A8/A9 ELISA

The S100A8/A9 levels in the plasma, PBMC/neutrophil EXOs, and PBMC/neutrophil EXO-free supernatants were measured using a Human S100A8/A9 Heterodimer Immunoassay Kit (DS8900, R&D Systems, USA) according to the manufacturer's recommendations. Each sample was tested in duplicate.

2.10. Statistical analysis

We used GraphPad Prism 5 for statistical analysis of the data except the microarray data. An unpaired *t*-test and two-way ANOVA was used for numerical variables with a normal distribution. The Wilcoxon rank sum test was used for data with a non-normal distribution. The data are shown as the mean ± SEM. $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. The identification of EXOs

EXOs were validated by TEM, bead-based flow cytometry and DLS. These EXOs showed a typical cup-shaped morphology approximately 130 nm in size and was highly expressed the EXO-specific marker CD63 (Fig. 1a–c). Meanwhile, CD63/CD9 gene expression was upregulated in SSc plasma EXOs compared with NC (Fig. 1d).

3.2. SSc neutrophil EXOs inhibited the proliferation and migration of HDMECs

SSc EXOs from both plasma and neutrophils can suppress the proliferation of HDMECs (Fig. 2a and b). Meanwhile, SSc neutrophil EXOs reduced the number of HDMECs that crossed the transwell membrane (Fig. 2c). Similarly, the migration analysis revealed that neutrophil EXOs from SSc patients decreased the wound-healing percentage in HDMECs at 12-h and 24-h points, however, plasma EXOs from SSc patients increased the percentage at the 24-h point (Fig. 2d). Together, these data indicate that neutrophil EXOs from SSc patients decreased the proliferation and migration of HDMECs.

3.3. The expressionof EXOs S100A8/A9 -in SSc patients

Using microarray analysis (Supplementary methods), we identified a total of 2307 altered genes in SSc PBMCs (1023 genes were upregulated and 1284 genes were downregulated) (Supplementary Fig. 1a). We integrated these 1023 upregulated genes with the genes obtained from the EXO databases ExoCarta [http://www.exocarta.org/] and exoRBase [http://www.exorbase.org/], which show genes likely present in human EXOs, and found 287 (28.1%) genes can exit in EXOs (Supplementary Fig. 1b). Bioinformatics analysis suggested five of the top 10 enriched pathways were focused on angiogenesis, vascular function, endothelial proliferation and apoptosis (Supplementary Fig. 1c). Remarkably, nine of the S100 family members were among the 287 genes, including S100A4, S100A6, S100A8, S100A9, S100A10, S100A11, S100A12, S100P, and S100Z (Supplementary Fig. 1d). S100A8/S100A9 is mainly secreted by neutrophils under physiological conditions [18], which might play an important role in angiopathy [19]. Hence, we further investigated the role of EXOs S100A8/A9 in SSc.

On the membrane surface of EXOs from plasma, PBMCs and neutrophils, we detected the positive expression of S100A8/A9 by bead-based flow cytometry (Fig. 3a). In SSc PBMCs and plamsa EXOs, the gene expression of S100A8 and S100A9 were increased (Fig. 3b). The concentrations of S100A8/A9 in plasma, neutrophil EXOs and PBMC EXOs, were distinctly higher in SSc than NC. Particularly, the S100A8/A9 concentration of neutrophil EXOs was higher than that in PBMC EXOs in SSc patients. While there were no differences in S100A8/A9 protein levels between neutrophil EXO-free supernatants and PBMC EXO-free supernatants (Fig. 3c). These results indicated that neutrophil EXOs were likely to be an important source of the elevated S100A8/A9 in peripheral blood observed in SSc patients.

3.4. SSc neutrophil EXOs S100A8/A9 inhibited the proliferation and migration of HDMECs

To investigate the effect of SSc neutrophil EXOs S100A8/A9 on HDMECs, we used the S100A8/A9 and S100A8 proteins as positive controls, and S100A8 antibody as inhibitors. We found that both the S100A8/A9 protein (500 ng/ml) and the S100A8 protein (100 ng/ml) decreased the proliferation activity of HDMECs compared with neutrophil EXOs (average concentration of S100A8/A9 around 10 ng/ml) (Fig. 4a). The S100A8 protein also reduced the number of transmembrane HDMECs, while the S100A8 antibody reversed and increased it (Fig. 4b). Similarly, the S100A8 antibody had a tendency to increase the number of transmembrane HDMECs co-stimulated with neutrophil EXOs from SSc patients (Fig. 4c). In accordance with these observations, SSc neutrophil EXOs S100A8/A9 inhibited the proliferation and migration of HDMECs.

4. Discussion

Our study firstly investigated the role of plasma and neutrophil EXOs in the vascular pathology of SSc, EXOs from SSc patients can inhibit the proliferation and migration of HDMECs. As S100A8/A9 can exist on the outside membrane of EXOs, further, we studied the function of EXOs S100A8/A9 in HDMECs. We found that S100A8/A9 expressed in neutrophil EXOs from SSc patients was likely to be an important source of the elevated S100A8/A9 in peripheral blood, however, more study is needed.

S100A8 and S100A9 (also named calgranulin A and calgranulin

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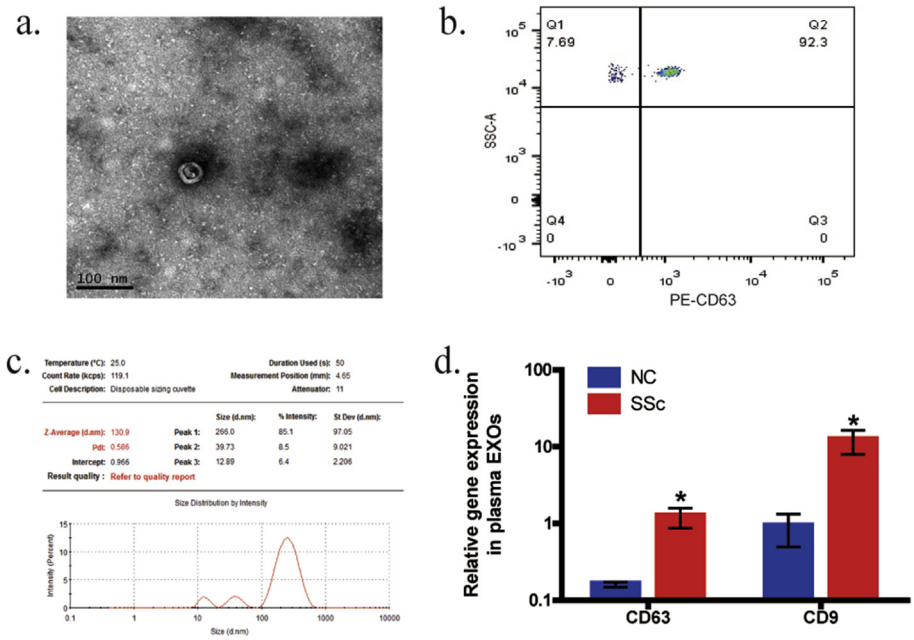

Fig. 1. The identification of EXOs. (a) EXOs showed the typical cup shape with TEM, (b) positive expression for CD63 in flow cytometry and (c) a size of approximately 130 nm by DLS. (d) CD63/CD9 gene expression was upregulated in SSc plasma EXOs. (*SSc vs NC).

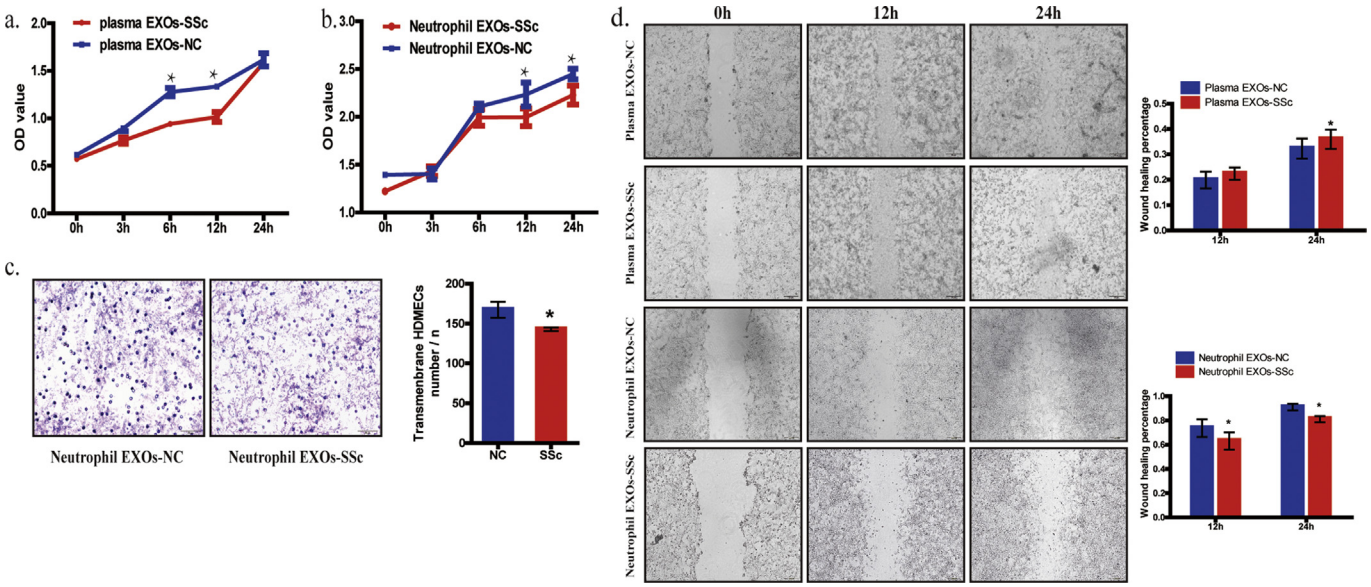

Fig. 2. SSc neutrophil EXOs decreased the proliferation and migration of HDMECs. (a) plasma EXOs and (b) neutrophil EXOs in SSc patients suppressed the proliferation of HDMECs. Neutrophil EXOs in SSc patients decreased (c) the number of transmembrane HDMECs and (d) wound-healing percentages at the 12-h and 24-h time point. (*SSc vs NC).

B), two S100 calcium-binding protein family members, mainly exist as the heterodimeric complex form of S100A8/A9 and bind specifically to endothelial cells, which represented approximately 60% of the heterodimeric S100A8/A9 in neutrophils under physiological conditions. Evidence indicates that S100A8/A9 secretion depends on an intact microtubule network or occurs in the form of vesicle secretion [18]. Futhermore, it has been proposed that S100A8/A9 might inhibit cell proliferation and differentiation, or induce cell death in myoblasts [19], colon cancer cells [20], leukemia cells [21] and normal fibroblasts [22]. In recent studies, the increase of endothelial permeability, and the loss of monolayer integrity and cell junction proteins after S1008 and S100A9 treatment had been shown in HDMECs and human umbilical venous endothelial cells, and those can be abrogated by preabsorption of S1008 and S100A9 with polyclonal anti-S1008 and anti-S100A9 antibodies [23,24]. Also, they demonstrated that S100A8/A9 can induce alteration indicative of both apoptotic and necrotic cell death, representing an important pathomechanism involved in vasculitis and other inflammatory diseases associated with endothelial dysfunction and accumulation of S100A8/A [25].

According to previous reports, S100A8/A9 is highly expressed in peripheral blood RNA [26], plasma [27], saliva [28], bronchoalveolar lavage fluid [29], and the supernatant of plasmacytoid dendritic cells in SSc patients [27,30]. The levels of S100A8/A9 have also been

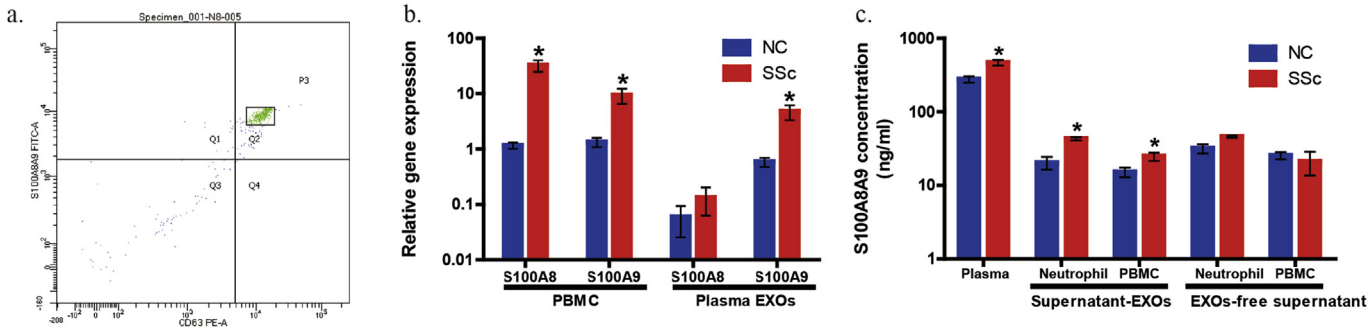

Fig. 3. The expression of EXOs S100A8/A9in SSc patients. (a) The positive expression of S100A8/A9 was detected on the membrane surface of EXOs from plasma, PBMCs and neutrophils. (b) S100A8 and S100A9 gene expression levels of PBMCs and plasma EXOs in SSc, were obviously higher than those in NC. (c) S100A8/A9 protein concentration in plasma, neutrophil EXOs and PBMC EXOs, were distinctly higher in SSc than that in NC. (*SSc vs NC).

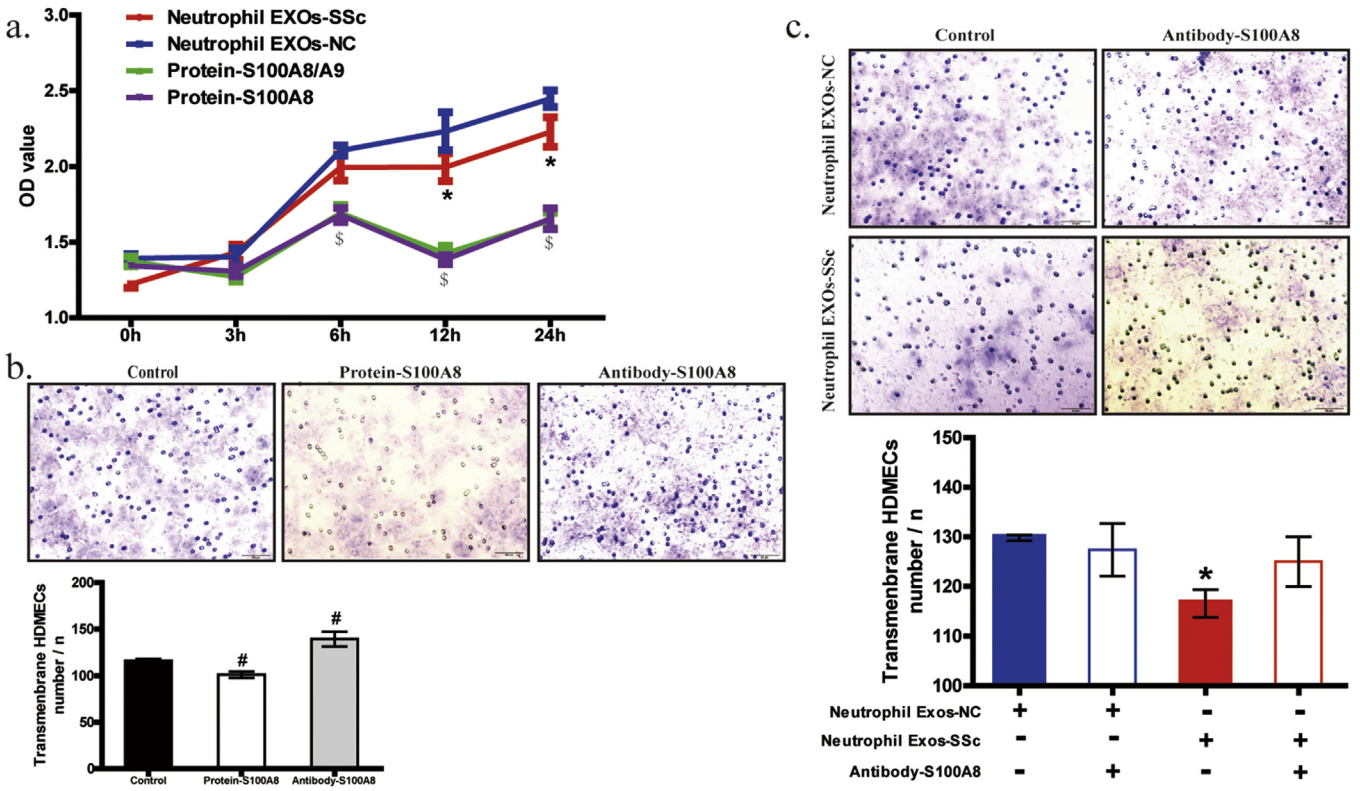

Fig. 4. SSc neutrophil EXOs S100A8/A9 inhibited the proliferation and migration of HDMECs. (a) The S100A8/A9 protein and S100A8 protein decreased the proliferation activity of HDMECs compared to neutrophil EXOs. (b) The S100A8 protein reduced the number of transmembrane HDMECs, while the S100A8 antibody reversed and increased the number of transmembrane cells. (c) The S100A8 antibody had a tendency to increase the number of transmembrane HDMECs stimulated by neutrophil EXOs from SSc patients. Each independent experiment was performed three times with duplicates in each group. (*SSc vs NC, $ SSc vs S100A8/A9 or S100A8, # S100A8 or the S100A8 antibody vs Control).

shown to be positively correlated with SSc-associated organ involvement and immunological abnormalities [26–29,31]. Studies on dermal fibroblasts in SSc patients have shown that S100A8/A9 may be the key molecule contributing to SSc fibrosis and inflammation [32,33]. Paquinimod, a small-molecule inhibitor of S100A8/A9 that blocks S100A8/A9 activity, abrogated the increase in autoantibody production in an SSc animal model [34,35]. Here, we found that S100A8/A9 not only existed in its dissociated form in plasma but was also located on EXOs, potentially on the membrane surface. Plasma S100A8/A9 may exert its functions by activating vascular endothelial cells or blood cells. However, due to the secretory and transport capacities of EXOs, EXO S100A8/A9 can be carried and delivered to target cells anywhere in the living body, thereby participating in many processes such as chemotaxis, inflammation, vascular reactivity and tissue repair [36].

In the pathogenesis of angiogenesis in autoimmune diseases, EXOs might act in the differentiation of endothelial progenitors, such as the proliferation and migration of endothelial cells [37]. This hypothesis was confirmed by our results showing that SSc neutrophil EXOs inhibited the proliferation and migration of HDMECs. On the other hand, in the immune response, EXOs may represent sources of self-antigens and play roles in immune complex formation, autoantigen presentation and immune suppression, potentially explaining the pathophysiological process of EXOs in SSc [38]. In next research, we would investigate the specific molecular mechanisms of neutrophil EXOs S100A8/A9 contributing

to inhibit angiogenesis in SSc.

5. Conclusion

Our study showed higher levels of S100A8/A9 in plasma, PBMCs, plasma EXOs, PBMC EXOs and neutrophil EXOs in SSc patients than that in NC. Most importantly, neutrophil EXOs has the angiogenesis inhibitory effects, and these changes were proved further by that neutrophil EXOs S100A8/A9 regulates vascular endothelial cell proliferation and migration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2020.03.088.

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